

Positive and negative numbers on the coordinate line

One line, one origin, one unit — and every integer has its own place.

LESSON 19–20

2 × 40 MIN

MATEMATIKA 7, §7

S8 1

LESSON CLOCK

15'

25'

25'

20'

5'

Building the line	15 min
Marking and reading	25 min
Comparing	25 min
Distance	20 min
Homework	5 min

LEARNING OBJECTIVES

- Draw a coordinate line with an origin, a direction and a unit.
- Mark integers and simple fractions on it.
- Read the coordinate of a marked point.
- Use the line to compare numbers and to find distances.

TEXTBOOK

National	pp. 47–52
Cambridge	Stage 8, pp. 2–10
Workbook	Exercise 1.1

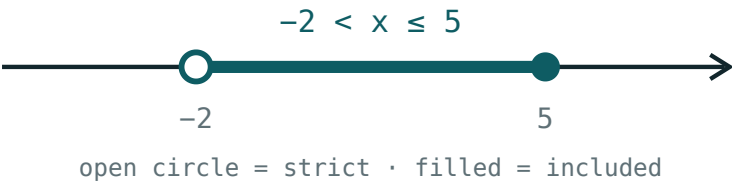
01

Explanation

Building the line

A **coordinate line** needs three things:

INGREDIENT	WHAT IT FIXES
an origin O	where 0 is
a positive direction	which way the numbers grow
a unit segment	how far apart consecutive integers are



An origin, a direction and a unit — and every number has a place.

The number attached to a point is its **coordinate**, written $A(-3)$.

EVERY POINT HAS EXACTLY ONE COORDINATE, AND EVERY NUMBER EXACTLY ONE POINT

That one-to-one correspondence is what makes the line useful: a question about numbers becomes a question about positions, and the other way round.

Marking and reading

NUMBER	POSITION
3	3 units right of O
-3	3 units left of O
0	at O
2.5	halfway between 2 and 3
-1.5	halfway between -1 and -2

-1.5 LIES BETWEEN -1 AND -2

Moving left from -1 the numbers become more negative. Reading the negative half as if it were the positive half reversed is the commonest error at this stage.

Comparing

Of two numbers, the one further to the right is the greater

COMPARISON	TRUE STATEMENT	WHY
3 and 5	$3 < 5$	5 is further right
-3 and 5	$-3 < 5$	any negative is less than any positive
-3 and -5	$-5 < -3$	-5 is further left
0 and -2	$-2 < 0$	zero beats every negative

AMONG NEGATIVES, THE ORDER IS REVERSED

$5 > 3$ but $-5 < -3$. The picture makes it obvious; the arithmetic alone does not. Sketching the line before comparing is worth the four seconds it costs.

Distance

The distance between two points is the number of units between them — always positive.

$$AB = |a - b|$$

POINTS	DISTANCE
3 and 7	4
-3 and 5	8
-7 and -2	5
-4 and 0	4

The second row is worth checking on a sketch: from -3 to 0 is 3 units, and from 0 to 5 is 5 more — 8 in all.

SUBTRACT, THEN TAKE THE SIZE
 $3 - 7 = -4$, but the distance is 4. A distance is never negative, whichever way the subtraction is done.

02 Worked examples

EXAMPLE 1 Mark -4 , -1.5 , 0 , 2 , 3.5 on a coordinate line and list them in ascending order.

- 1 Draw the origin, the direction and the unit.
- 2 -4 is four units left; -1.5 is between -1 and -2 .
- 3 Reading left to right gives the order.
- 4 $-4 < -1.5 < 0 < 2 < 3.5$

ANSWER

$-4, -1.5, 0, 2, 3.5$

EXAMPLE 2 Find the distance between the points with coordinates -3 and 5 .

- 1 $|-3 - 5| = |-8|$
- 2 $= 8$
- 3 Check on the line: 3 units to zero, 5 more.
- 4 8 units.

ANSWER

8

EXAMPLE 3 Which is greater, -7 or -2 ? And -7 or 0.5 ?

① -2 is further right than -7 .

② $-7 < -2$

③ Any negative is less than any positive.

④ $-7 < 0.5$

ANSWER

-2 and 0.5

03 Interactive model

Chalk a number line along the floor and have pupils stand on given coordinates; comparing and distance become physical before they are symbolic.

The coordinate line

$$x > 2$$



inequality $x > 2$

boundary **2 (open)**

$x = 0$ works? **no**

$x = 6$ works? **yes**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Coordinate line	Koordinata to'g'ri chizig'i	Координатная прямая
Origin	Boshlanish nuqtasi	Начало отсчёта
Unit segment	Birlik kesma	Единичный отрезок
Coordinate	Koordinata	Координата
Positive direction	Musbat yo'nalish	Положительное направление
To the right	O'ngda	Справа
To the left	Chapda	Слева
Distance	Masofa	Расстояние

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A coordinate line needs:

an origin only

an origin, a direction and a unit

a unit only

two points

-1.5 lies between:

-1 and 0

-1 and -2

1 and 2

0 and 1

The greater of -3 and -5 :

-3

-5

equal

cannot tell

Any negative number is:

greater than every positive

less than every positive

equal to zero

undefined

The distance between -3 and 5 :

2

8

-8

15

A distance is:

sometimes negative

never negative

always zero

a coordinate

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Greater: 3 or 5

ANSWER 5

2. Greater: -3 or 5

ANSWER 5

3. Greater: -3 or -5

ANSWER -3

4. Greater: 0 or -2

ANSWER 0

5. Distance from 3 to 7

ANSWER 4

6. Distance from -7 to -2

ANSWER 5

7. Distance from -4 to 0

ANSWER 4

Medium · the standard question

7 PROBLEMS

1. Ascending: $-4, -1.5, 0, 2, 3.5$

ANSWER As written

2. Ascending: $2, -7, 0, -1, 5$

ANSWER $-7, -1, 0, 2, 5$

3. Distance from -3 to 5

ANSWER 8

4. Distance from -12 to 4

ANSWER 16

5. The point halfway between -6 and 2

ANSWER -2

6. The point halfway between -9 and -1

ANSWER -5

7. Descending: $-2, -8, 3, 0$

ANSWER $3, 0, -2, -8$

Hard · stretch and reason

7 PROBLEMS

1. All integers x with $-4 < x \leq 2$

ANSWER $-3, -2, -1, 0, 1, 2$

2. How many integers between -15 and 9 ?

ANSWER 23

3. Points 5 units from -2

ANSWER 3 and -7

4. A point twice as far from 0 as -4 , on the same side

ANSWER -8

5. The midpoint of a and b

ANSWER $\frac{a+b}{2}$

6. Distance between a and $-a$

ANSWER $2|a|$

7. All x with distance 3 from 7

ANSWER 4 and 10

07 Homework

Homework — 5 tasks

Sketch the line for every comparison; the picture is the argument.

1. Mark $-5, -2.5, 0, 1, 4$ on a coordinate line.
2. Put in ascending order: $3, -6, 0, -1, 2$.
3. Find the distance between -8 and 3 .
4. Find the point halfway between -10 and 4 .

5. List all integers x with $-5 \leq x < 3$.